



CCY2001 Introduction to Cybersecurity - Make-up Exam (30 Points)

Student ID: _____ Student Name: _____ Class: _____

Question One: Multiple Choice Questions (10 points)

1. In the context of the CIA triad, which property is violated if an attacker executes a *Distributed Denial of Service (DDoS)* attack against a web server?
 - (a) Confidentiality
 - (b) Integrity
 - (c) Availability
 - (d) Authenticity
2. Which of the following best describes a **Passive Attack**?
 - (a) The attacker intercepts data to read the contents but does not alter it.
 - (b) The attacker modifies the data in transit to change its meaning.
 - (c) The attacker impersonates a legitimate user to gain access to a system.
 - (d) The attacker floods a system with requests to shut it down.
3. Which type of malware is specifically designed to encrypt a user's files and demand payment for the decryption key?
 - (a) Spyware
 - (b) Ransomware
 - (c) Logic Bomb
 - (d) Rootkit
4. In a **Man-in-the-Middle (MitM)** attack, the attacker primarily targets which aspect of the CIA triad?
 - (a) Only Availability
 - (b) Confidentiality and Integrity
 - (c) Physical security of the hardware
 - (d) Storage capacity of the database
5. What is the primary purpose of **Salting** a password before hashing it?
 - (a) To make the password shorter for faster processing.
 - (b) To ensure that two users with the same password have different hashes.
 - (c) To allow the server to decrypt the password more easily.
 - (d) To replace the need for multi-factor authentication.

Scenario: A software developer notices that a web application allows users to submit comments that include HTML and Script tags. An attacker submits a comment containing a script that redirects all visiting users to a fake login page designed to steal credentials. While the server remains online, the database now contains unauthorized malicious scripts.

- 1: This scenario is a classic example of which type of vulnerability?
 - (a) SQL Injection
 - (b) Cross-Site Scripting (XSS)
 - (c) Zero-Day Exploit
 - (d) Buffer Overflow
- 2: Which security principle was primarily violated when the malicious script was stored in the database?
 - (a) Availability
 - (b) Accountability
 - (c) Integrity
 - (d) Non-repudiation

- 3: If the developer implements **Input Validation**, which stage of the attack is they trying to prevent?
- The initial injection of the code
 - The user clicking the link
 - The redirection to the fake page
 - The stealing of the cookies
- 4: In this scenario, the "Fake Login Page" is a tool used for:
- Social Engineering / Phishing
 - Dictionary Attack
 - Brute Force
 - Passive Eavesdropping
- 5: What is the **Impact** of this attack on the end-users?
- Their computers will physically break.
 - Their account credentials may be compromised.
 - The web server will experience a hardware failure.
 - The internet speed of the user will increase.

Question Two: Essay Question (20 points)

Alphabet to Numerical Code Mapping

Letter	a	b	c	d	e	f	g	h	i	j	k	l	m
Code	0	1	2	3	4	5	6	7	8	9	10	11	12

Letter	n	o	p	q	r	s	t	u	v	w	x	y	z
Code	13	14	15	16	17	18	19	20	21	22	23	24	25

- 1- Encrypt the word **cyberdefense** using the Vigenère cipher with the key **key**.
- 2- Using the Row Transposition Cipher with key (3 1 5 2 4), encrypt the following text: **The data is encrypted now.**
- 3- Suppose that the plaintext **help** is encrypted to the ciphertext **Hiqy** using the Hill cipher (2x2 matrix); find the encryption key.
- 4- Using Playfair cipher with a key (**cybersecurity**), construct its matrix then encrypt the following sentence: **attack at dawn.**
- 5- Decrypt the ciphertext **"fshj"** using the function D defined by $D(C) = D_2(D_1(C))$, where:
 - D_1 is the decryption of the Vigenère cipher with key (3, 7, 11).
 - D_2 is the decryption of the substitution cipher E_2 provided below.

Substitution Table for E_2

E_2	a	b	c	d	e	f	g	h	i	j	k	l	m
Cipher	q	w	e	r	t	y	u	i	o	p	a	s	d

E_2	n	o	p	q	r	s	t	u	v	w	x	y	z
Cipher	f	g	h	j	k	l	z	x	c	v	b	n	m